

HYPOTHERMIA MANAGEMENT

Defined as core temperature $<35^{\circ}\text{C}$. Hypothermia cardiac arrest management differs from normal ALS algorithms. FCI & NFI's are NOT in this guideline - refer to JSP 375 Chp 42.

FULL PRIMARY SURVEY (ABCDE) and CORE TEMPERATURE FOR SEVERITY

MILD		MODERATE		SEVERE	
TEMP	32-35°C	TEMP	28-32°C	TEMP	$<28^{\circ}\text{C}$
AVPU ⁽¹⁾	ALERT/ORIENTATED	AVPU ⁽¹⁾	VOICE or CONFUSED	AVPU ⁽¹⁾	UNRESPONSIVE
SIGNS	SHIVERING	SIGNS	+/- SHIVERING	SIGNS	SLOW VITALS ⁽²⁾

PREVENT FURTHER HEAT LOSS and ASSESS PATIENT

BREATHING	RATE (10 to 30), SPO ₂ (94-98%), ETCO ₂ (4.0 to 5.7kPa)
CIRCULATION	PULSE RATE/STRENGTH & BP (SHOCK), CONTINUOUS ECG FOR ARRHYTHMIA ⁽³⁾
DISABILITY	AVPU or GCS, BLOOD SUGAR AND CONTINUOUS RECTAL TEMP MONITORING

REMOVE WET CLOTHING, GET PATIENT OFF THE GROUND AND INTO SLEEPING BAG WRAPPED WITH HPMK AND HAT
USE APPROPRIATE SHELTER IN WET AND WINDY ENVIRONMENTS, IF UNAVAILABLE USE WATERPROOF COVER

TREATMENT - REWARMING METHODS

MILD	PROVIDE WARM SUGARY DRINKS IF CONSCIOUS AND CLOTH WRAPPED WARMING PACKS OR BOTTLES WITH WARM WATER (42°C) INTO AXILLA AND GROIN
MODERATE	SAME AS ABOVE, CONSIDER IV ACCESS AND REGULAR MONITORING
SEVERE	PREPARE FOR POTENTIAL HYPOTHERMIC CARDIAC ARREST WITH FULL MONITORING
AIRWAY	P or U = REDUCED AIRWAY REFLEXES - MANAGE WITH ADJUNCTS
BREATHING	MAINTAIN RATE >10 , MAINTAIN SPO ₂ 94-98%, MAINTAIN ETCO ₂ (4.0 to 5.7kPa)
CIRCULATION	IF REDUCED CONSCIOUSNESS: GAIN IV/IO ACCESS, GIVE BOLUSES OF WARM (40°C) SALINE OR USE A FLUID WARMER. CONSIDER GLUCOSE 10% FOR HYPOGLYCAEMIA

COMPLICATIONS DURING HANDLING, TREATMENT AND TRANSPORT

AFTERDROP	CORE TEMPERATURE DROPS DURING REWARMING - CONTINUE REWARMING METHODS AND CONSIDER WARM IV FLUIDS.
HYDROSTATIC PRESSURE	WHEN REMOVING PATIENTS FROM PROLONGED WATER IMMERSION, REMOVE FROM WATER HORIZONTALLY TO AVOID SIGNIFICANT HYPOTENSION FROM VENOUS POOLING.
TRANSPORT AND EVAC	TRANSPORT HORIZONTALLY TO PROMOTE VENOUS RETURN, SITE AWAY FROM COLD PARTS OF AIRCRAFT AND REQUEST IF HEATING CAN BE TURNED ON

HYPOTHERMIC CARDIAC ARREST MANAGEMENT

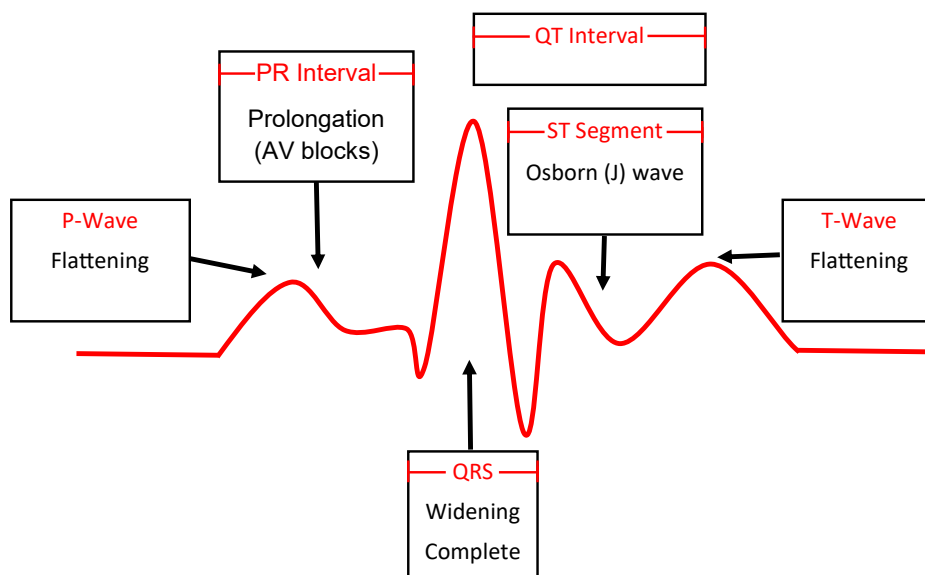
RECOGNITION	COMMENCE CPR IF INEFFECTIVE OR NOT BREATHING PULSES ARE UNRELIABLE IN HYPOTHERMIC PATIENTS (USE ECG and ETCO ₂ MONITORING)
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ONLY SHOCK PATIENT ONCE UNTIL CORE TEMPERATURE $>30^{\circ}\text{C}$.

IF THE CHEST IS COMPLETELY STIFF THEN DO NOT COMMENCE CPR

DO NOT ADMINISTER DRUGS $<30^{\circ}\text{C}$. DOUBLE DRUG INTERVAL TIMES IF 30-35°C.

CONSIDER PROLONGED RESUSCITATION TO ROLE 2 IN ICY COLD WATER IMMERSION PTs.
TERMINATION SHOULD BE CONSIDERED IF POTASSIUM IS $>12\text{mmol/L}$.
CONSIDER TRANSPORTING TO MTF WITH ECMO CAPABILITY



POINTS TO NOTE

1. CNS DYSFUNCTION IS THE BIGGEST INDICATOR OF SEVERITY, ESPECIALLY AS TYMPANIC TEMPERATURE MONITORS CAN BE INACCURATE. CLINICAL JUDGEMENT SHOULD INCLUDE ENVIRONMENTAL FACTORS (IMMERSION OR ICY CONDITIONS) AND CLINICAL PRESENTATION.
2. VITAL SIGNS SUCH AS HEART RATE, BREATHING RATE AND GCS COULD DECREASE AS HYPOTHERMIA PROGRESSES. SPO₂ MAY BE UNREADABLE DUE TO COLD PERIPHERIES. ALWAYS CHECK BLOOD SUGAR IN REDUCED CONSCIOUSNESS AND TREAT ACCORDINGLY.
3. ARRHYTHMIA IS LIKELY TO BE BRADYCARDIA AND CAN PRESENT WITH OSBOURNE (J) WAVES, PREMATURE ATRIAL OR VENTRICULAR CONTRACTIONS, OR PROLONGED QRS OR QT INTERVAL.
4. AVOID PUTTING CHEMICAL HAND WARMERS ONTO BARE SKIN AS IT COULD CAUSE BURNS.
5. ENERGY STORES CAN BE DEPLETED WITH HOURS SO SUPPLEMENTS ARE ADVISED TO MAINTAIN ENERGY FOR SHIVERING OR ACTIVE MOVEMENT TO PROMOTE WARMING IN FULLY CONSCIOUS PATIENTS. CONSIDER HYPOGLYCAEMIA IN HYPOTHERMIC PATIENTS.
6. AFTERDROP IS WHERE CORE TEMPERATURE CONTINUES TO DROP AFTER REMOVAL FROM COLD STRESS. CAUSE IS A COMBINATION OF CONDUCTIVE HEAT LOSS WHEN WARM CORE BLOOD IS DISSEMINATED TO COOL PERIPHERAL TISSUE AND SUBSEQUENT COOLER BLOOD RETURNING TO CENTRAL CIRCULATION AND HEART.
7. WHEN REMOVING PROLONGED IMMERSED PATIENTS, LOSS OF HYDROSTATIC PRESSURE CAN RESULT IN VENOUS BLOOD POOLING WITH GRAVITY, SO IT IS ADVISED TO REMOVE PATIENTS HORIZONTALLY AND MAINTAIN THAT POSITION UNTIL CARDIOVASCULAR STATUS IMPROVES.
8. THERE'S AN INCREASED RATE OF GOOD NEUROLOGICAL OUTCOME FOLLOWING PROLONGED HYPOTHERMIC CARDIAC ARREST IN PATIENTS THAT WERE EXPOSED TO CONDITIONS THAT CAUSED A RAPID DECLINE IN CORE TEMPERATURES, SUCH AS ICY WATER IMMERSION.

Guideline written in accordance with SFM Clinical Standard Operating Procedure 013 'Management of Hypothermia' (2022)

References

- Dow, J., et al. 2019 Wilderness Medical Society Clinical Practice Guidelines for the Out-of-Hospital Evaluation and Treatment of Accidental Hypothermia: 2019 Update, *Wilderness Environment Medicine*, 30(4), pp. S47-S69.
- Lott, C., et al. 2021. European Resuscitation Council Guidelines 2021: Cardiac Arrest in Special Circumstances, *Resuscitation*, 161, pp. 152-219. Available at: [European Resuscitation Council Guidelines 2021: Cardiac arrest in special circumstances - Resuscitation \(resuscitationjournal.com\)](https://www.resuscitationjournal.com/article/S0950-2688(21)00152-1)